



صندوق
SANDOOQ

SANDOOQ

Sovereign, end-to-end encrypted cloud storage, owned by Saudis who use it

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QR CODE

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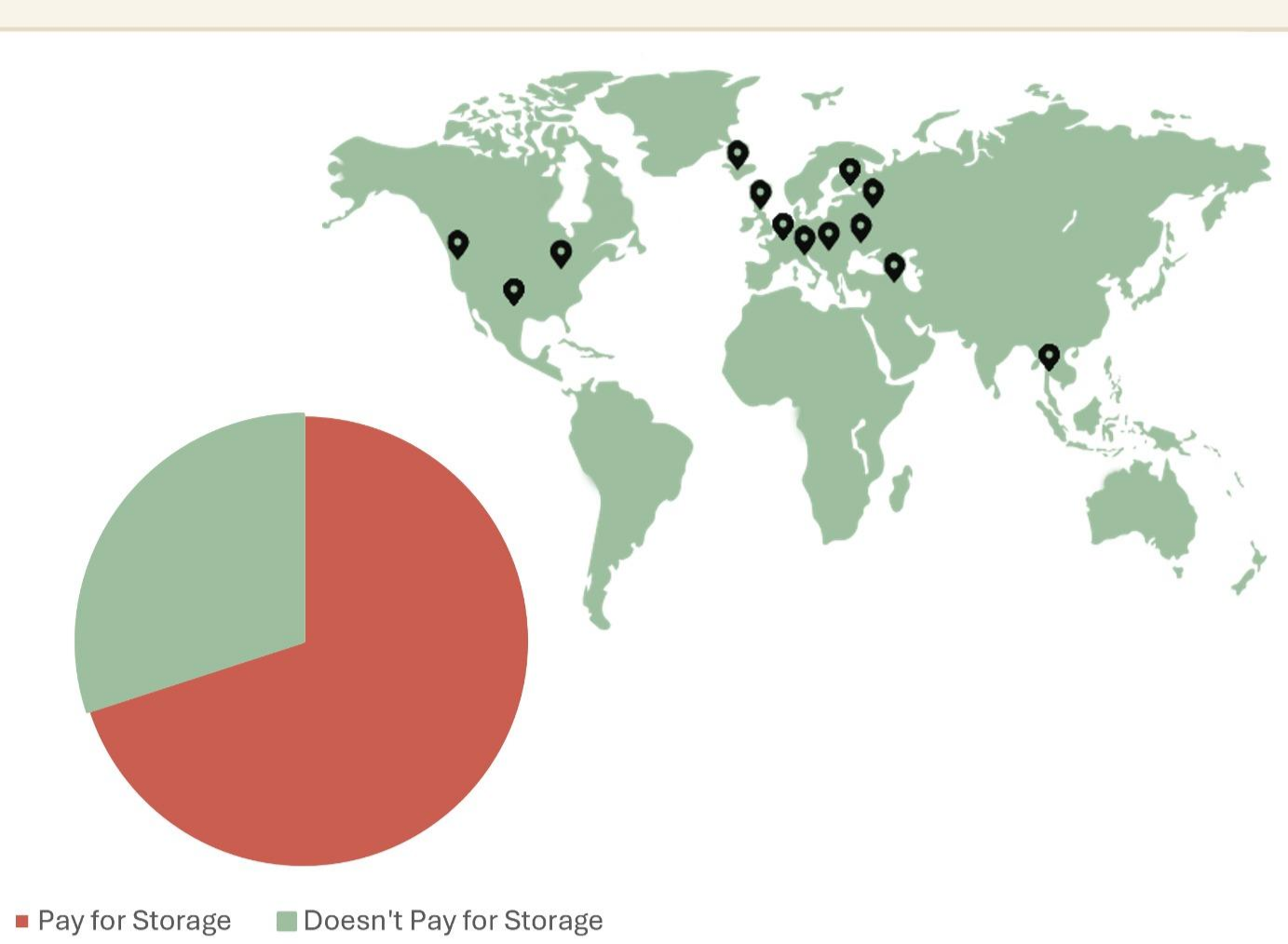
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Introduction



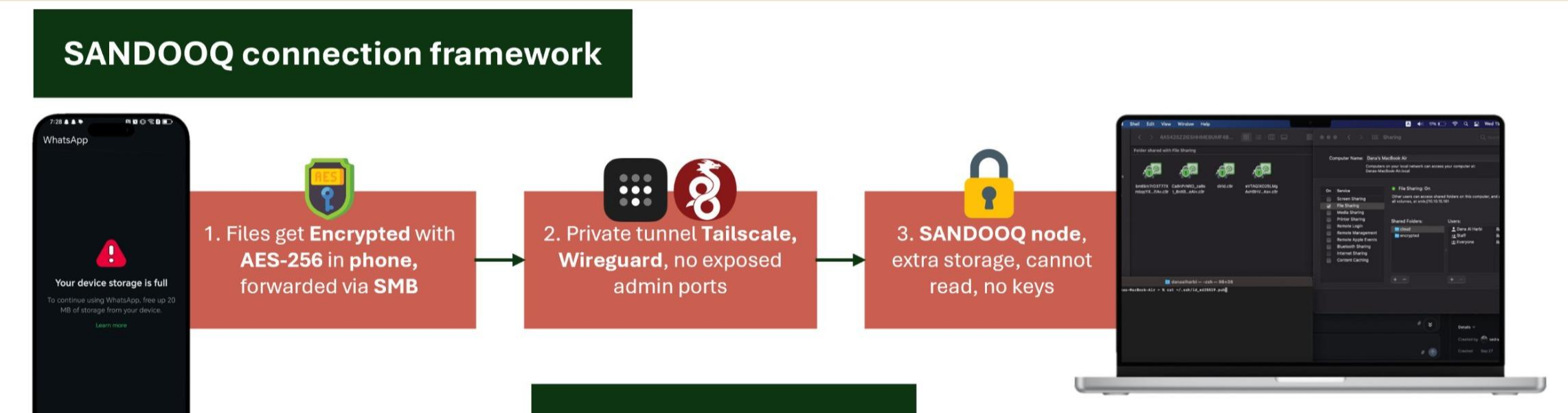


- 70% of iPhone users pay after running out of storage, \$70/year is the average payer spending . (1)
- 85% of data-center electricity consumption are based in US, Europe and China. Saudi data -that is expected to reach 20 exabytes, with 50+ million devices- mostly lives in foreign regions(3) (4)
- \$380.15 bn Global cloud storage projected market value by 2031 (2)
- SANDOOQ: Privacy by mathematics, not by policy. Every file is encrypted on the user’s device before it travels, operators cannot read, not even me.
- SANDOOQ: Sovereignty by design. Storage nodes live inside Saudi, aligned with CST/NCA requirements.
- SANDOOQ: A business, not just a solution. Personal Self-hosting or Subscriptions on demand, marketplace anyone with rentable spare storage.

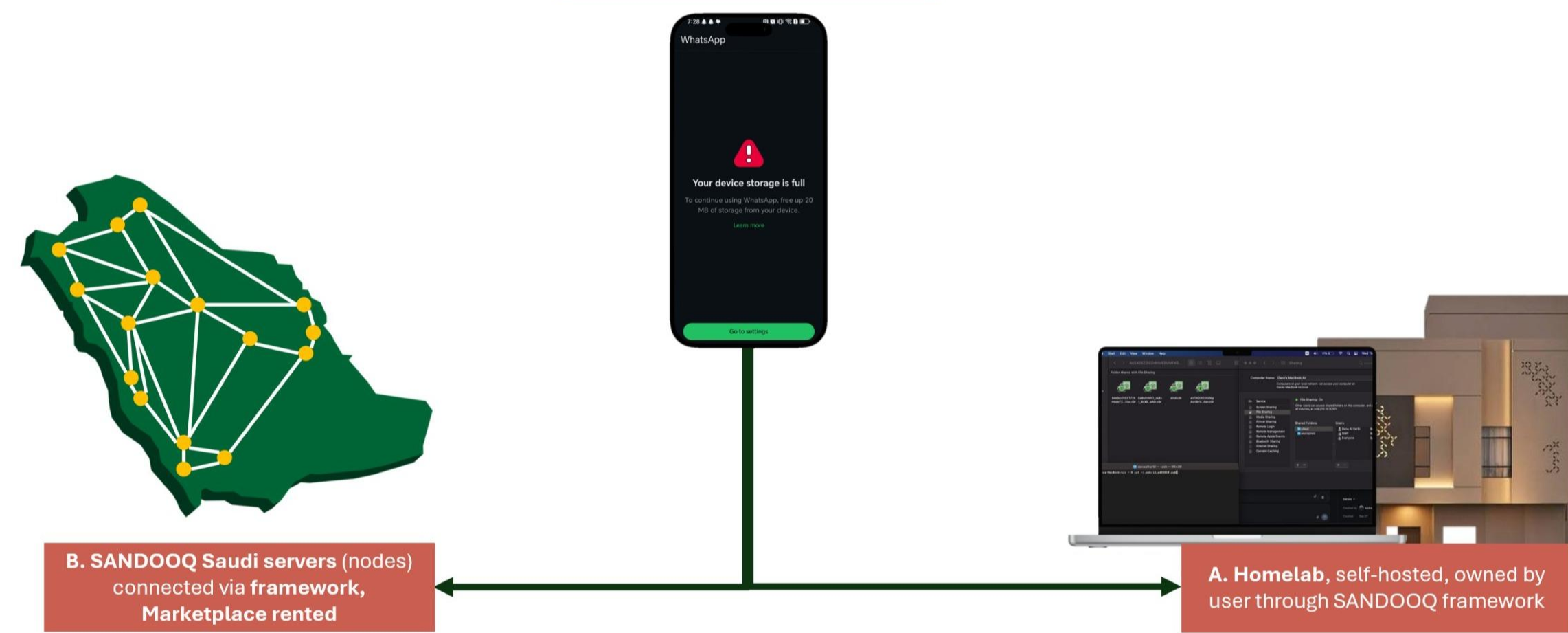
iCloud Storage is Full. Either delete something precius or pay again. This pain it is a business model, and it has become the default way the world stores its memories.

2

Methodology



SANDOOQ connection framework



SANDOOQ storage types

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Connection to server via SMB, user experience similar to icloud.

Selected files Encryption with AES-256 on mobile

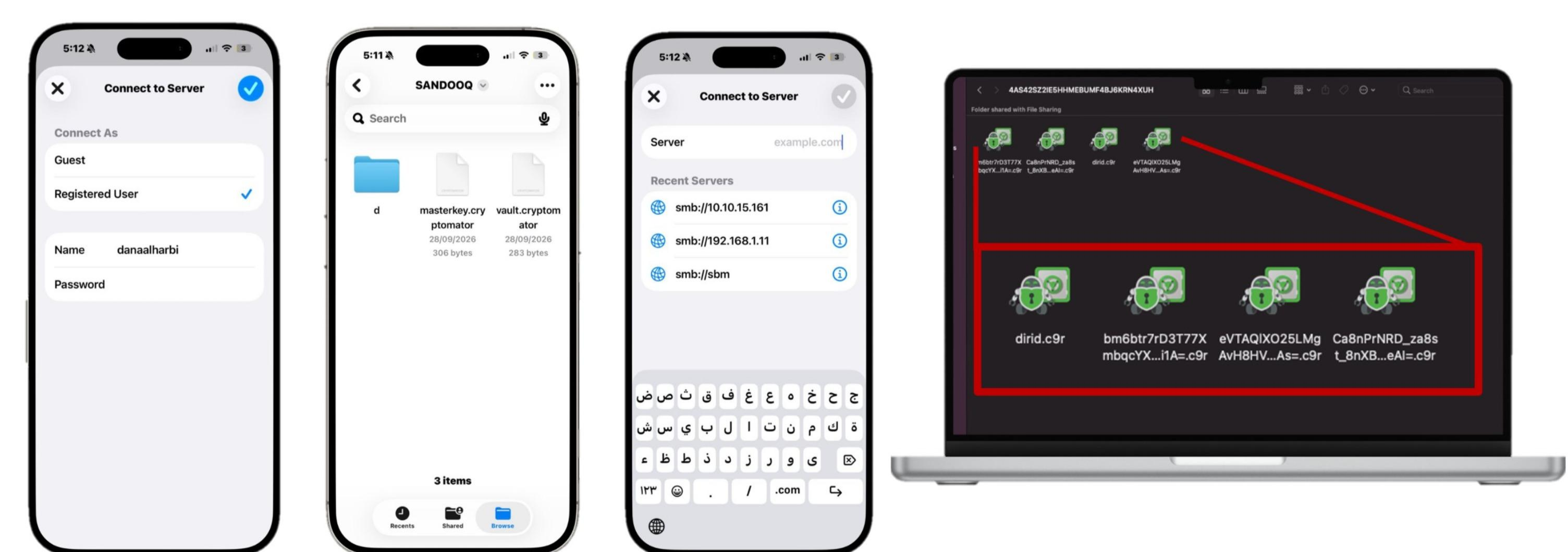
Files travel with private tunneling with Tailscale & Wireguard for privacy and Security

Selected server (home-server/ Saudi-server) receives files as blocks, encrypted with no keys

Mobile user always can access files as key lives in mobile, otherwise always encrypted

3

Results



2

0

Instant

Tested encrypted nodes on tailnet

readable bytes for the operator

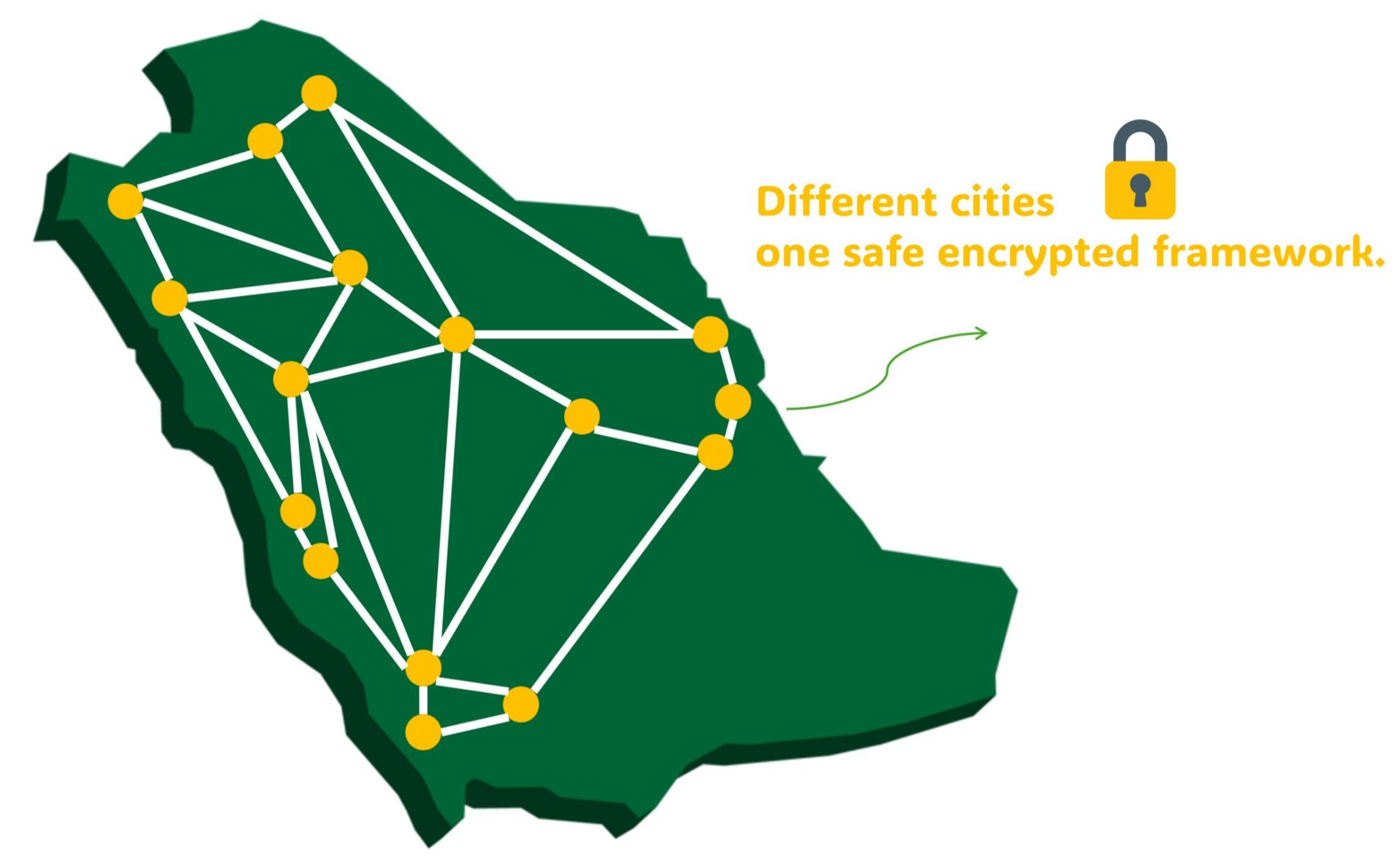
Real

- Working prototype across a public node and a home node via SMB.
- On-disk files verified unreadable; only encrypted .c9r names are visible.
- AES-256 Encrypted vault round-trip verified from phone and server.
- Verified more storage for the phone, no icloud subscription.

Acknowledgments Statement: My project received assistance and supervision from Numou Educaton Center, Creativity Oasis in Alkhobar, my supervisor Sedra Abdulbasir, and Okal.ai platform.

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Innovation



- Real-World Testing:** anyone can run a node and earn a monthly payout for unused space.
- Storage as local infrastructure:** anyone can run a node and earn a monthly payout for unused space.
- Sovereign by design:** hosted and administered inside the Kingdom.
- No exposed admin surface:** SSH is closed to the internet; operators use the private tailnet.

5

Conclusion



Encrypted on the device

Hosted in the Kingdom

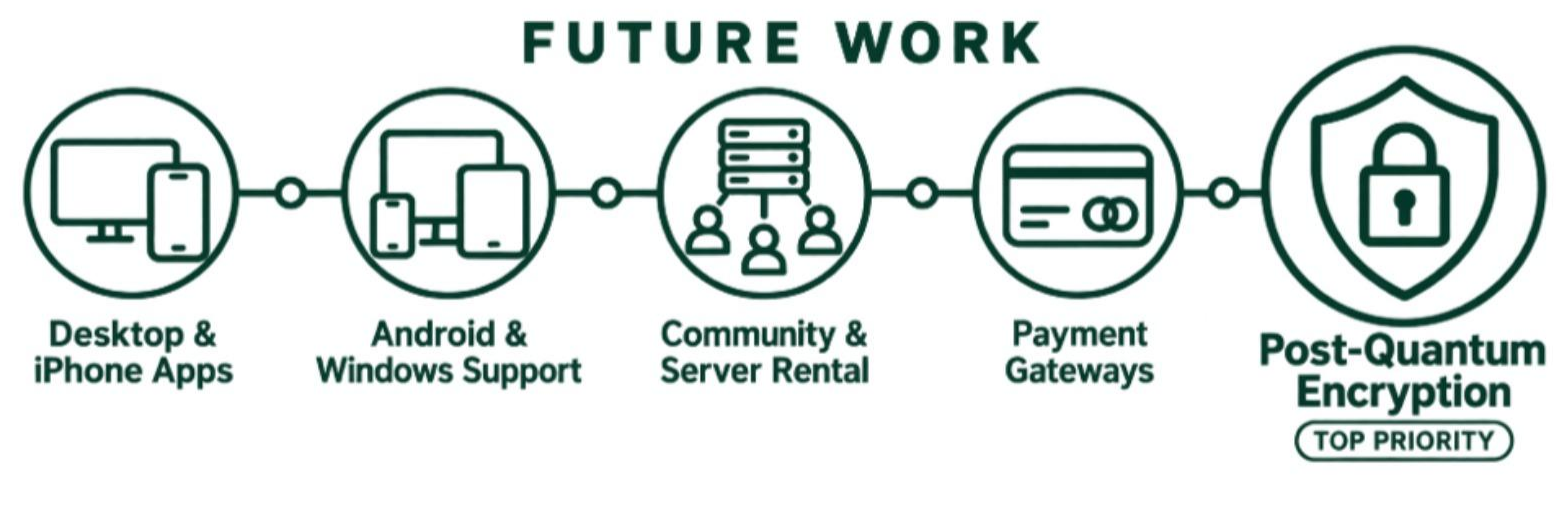
Saudis value privacy, we all should protect it

Supplied by the community

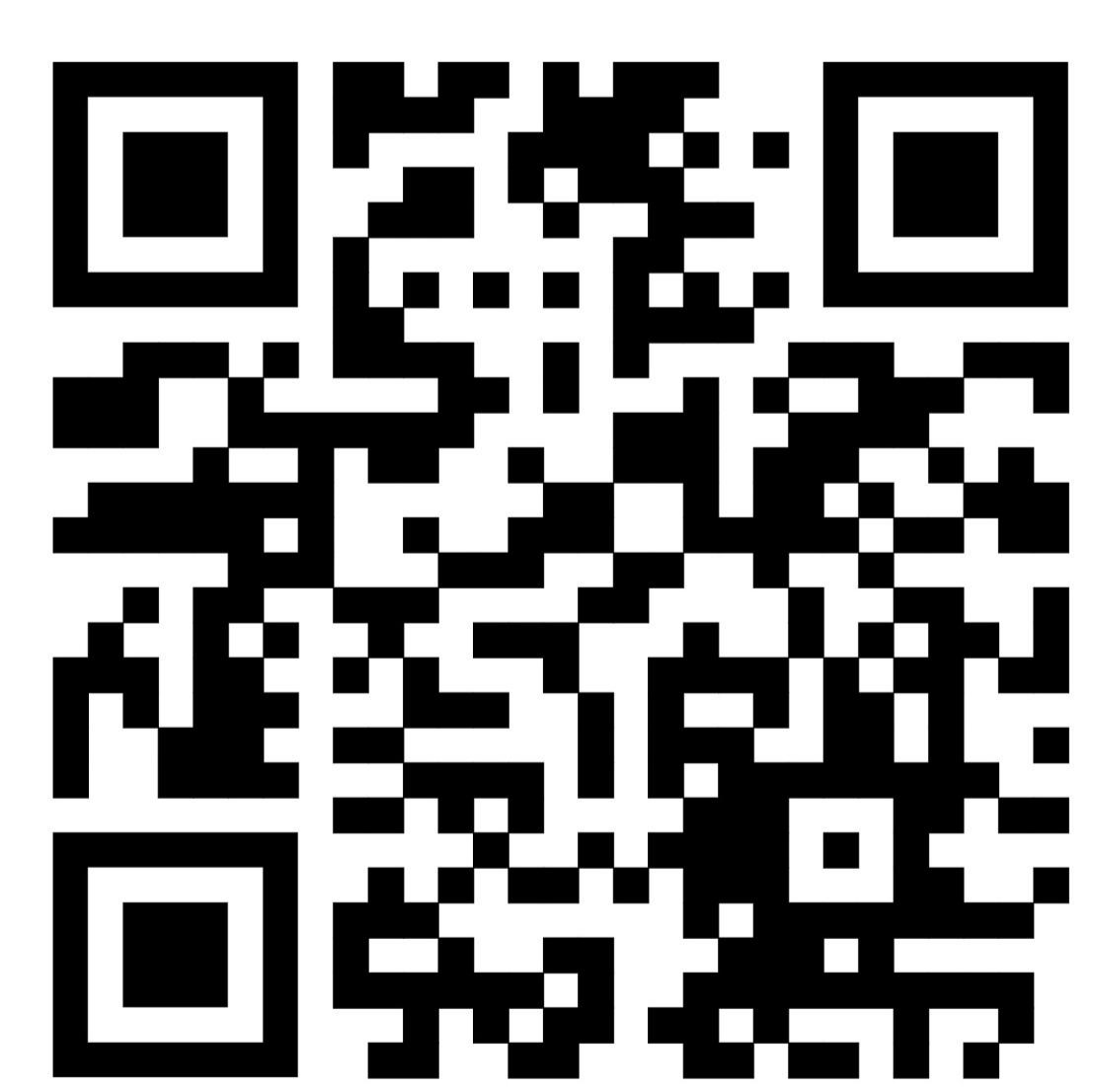
Storage you own. Privacy you can verify. Income that stays in the Kingdom.

6

Future Work & Reference



FUTURE WORK



- Second federated node and an in-app node directory.
- Minimum host earning flow: register capacity, accept encrypted writes.
- One-command installer so anyone can reproduce the setup.
- Post Quantum Encryption for future bullet proofing.

References: Scan QR Code